EXPERIMENT 1

ADF Pipeline: On-Prem SQL to Azure Blob Storage

Aim

To perform adf pipeline: on-prem sql to azure blob storage using Microsoft Azure services.

Theory

This experiment introduces the Azure services, architecture, workflow, advantages, industry applications, and implementation methodology. Students gain hands-on experience with cloud-native analytics solutions used in modern enterprises. Azure Data Factory (ADF) is a cloud-based data integration service that enables organizations to create, schedule, and manage data movement workflows. It supports data ingestion from multiple sources, including on-premises databases, cloud storage, and SaaS applications. In this experiment, ADF is used to transfer data from an on-premises SQL Server database to Azure Blob Storage using a Self-Hosted Integration Runtime. This demonstrates hybrid cloud data integration and forms the foundation of modern data engineering solutions.

Flow Diagram

On-Prem SQL Server

▼

Self-hosted Integration Runtime

▼

Azure Data Factory

▼

Azure Blob Storage

Procedure

1. Create Azure Data Factory.
2. Configure Self-hosted Integration Runtime.
3. Connect SQL Server as source.
4. Connect Azure Blob Storage as sink.
5. Create Copy Data Activity.
6. Run and monitor the pipeline.

Expected Output

<GitHub Link Here>

Result

Thus, the Azure Data Factory pipeline was successfully created to ingest data from an on-premises SQL Server database into Azure Blob Storage. This approach is widely used in industries for data migration, data warehousing, and cloud modernization projects. Organizations use Azure Data Factory to automate data movement from legacy systems to cloud platforms, enabling centralized data storage, improved scalability, and better support for analytics and reporting.